

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
Dataset found at:
/content/drive/MyDrive/Bone_Fracture_Binary_Classification
```

```
Using Device: cuda
GPU: Tesla T4
```

```
=====
DATASET INFORMATION
=====
```

```
Training samples   : 9246
Validation samples  : 829
Testing samples    : 506
Classes            : ['fractured', 'not fractured']
Class mapping      : {'fractured': 0, 'not fractured': 1}
```

```
=====
STARTING SEED 42
=====
```

```
Seed 42 | Epoch 1/10
Train Loss: 0.2799 | Val Loss: 0.1261
Train Acc: 89.90% | Val Acc: 94.69%
Epoch Time: 3.43 minutes
```

```
Seed 42 | Epoch 2/10
Train Loss: 0.0414 | Val Loss: 0.0583
Train Acc: 98.80% | Val Acc: 98.07%
Epoch Time: 1.01 minutes
```

```
Seed 42 | Epoch 3/10
Train Loss: 0.0213 | Val Loss: 0.0518
Train Acc: 99.41% | Val Acc: 98.19%
Epoch Time: 1.01 minutes
```

```
Seed 42 | Epoch 4/10
Train Loss: 0.0141 | Val Loss: 0.0469
Train Acc: 99.72% | Val Acc: 98.19%
Epoch Time: 1.03 minutes
```

```
Seed 42 | Epoch 5/10
Train Loss: 0.0119 | Val Loss: 0.0443
Train Acc: 99.73% | Val Acc: 98.19%
Epoch Time: 1.01 minutes
```

```
Seed 42 | Epoch 6/10
Train Loss: 0.0112 | Val Loss: 0.0420
Train Acc: 99.72% | Val Acc: 98.31%
Epoch Time: 1.00 minutes
```

```
Seed 42 | Epoch 7/10
Train Loss: 0.0085 | Val Loss: 0.0439
Train Acc: 99.79% | Val Acc: 98.19%
Epoch Time: 1.02 minutes
```

```
Seed 42 | Epoch 8/10
Train Loss: 0.0097 | Val Loss: 0.0410
Train Acc: 99.79% | Val Acc: 98.31%
Epoch Time: 1.00 minutes
```

```
Seed 42 | Epoch 9/10
Train Loss: 0.0086 | Val Loss: 0.0450
Train Acc: 99.84% | Val Acc: 98.19%
Epoch Time: 0.99 minutes
```

```
Seed 42 | Epoch 10/10
Train Loss: 0.0111 | Val Loss: 0.0412
Train Acc: 99.79% | Val Acc: 98.43%
Epoch Time: 1.00 minutes
```

```
=====
SEED 42 TRAINING COMPLETE
Time taken: 12.50 minutes
=====
```

Calculating final metrics for Seed 42...

=====

SEED 42 - FINAL PERFORMANCE

=====

Metric	Training	Validation	Testing
Accuracy (%)	99.9135	98.3112	97.6285
Sensitivity (%)	99.9353	98.7805	97.7612
Specificity (%)	99.8914	97.6261	97.4790
Precision (%)	99.8923	98.3806	97.7612
F1-Score (%)	99.9138	98.5801	97.7612
Error Rate (%)	0.0865	1.6888	2.3715
ROC-AUC	1.0000	0.9991	0.9987

Seed 42 - Training Confusion Matrix:
[[4601 5]
[3 4637]]

Seed 42 - Validation Confusion Matrix:
[[329 8]
[6 486]]

Seed 42 - Testing Confusion Matrix:
[[232 6]
[6 262]]

Completed 1/5 seeds.

=====

STARTING SEED 52

=====

Seed 52 | Epoch 1/10
Train Loss: 0.2911 | Val Loss: 0.1313
Train Acc: 88.71% | Val Acc: 93.73%
Epoch Time: 1.00 minutes

Seed 52 | Epoch 2/10
Train Loss: 0.0433 | Val Loss: 0.0760
Train Acc: 98.75% | Val Acc: 96.86%
Epoch Time: 1.00 minutes

Seed 52 | Epoch 3/10
Train Loss: 0.0239 | Val Loss: 0.0525
Train Acc: 99.33% | Val Acc: 98.31%
Epoch Time: 1.00 minutes

Seed 52 | Epoch 4/10
Train Loss: 0.0140 | Val Loss: 0.0509
Train Acc: 99.56% | Val Acc: 98.31%
Epoch Time: 1.00 minutes

Seed 52 | Epoch 5/10
Train Loss: 0.0147 | Val Loss: 0.0539
Train Acc: 99.58% | Val Acc: 98.31%
Epoch Time: 1.00 minutes

Seed 52 | Epoch 6/10
Train Loss: 0.0126 | Val Loss: 0.0513
Train Acc: 99.61% | Val Acc: 97.95%
Epoch Time: 1.00 minutes

Seed 52 | Epoch 7/10
Train Loss: 0.0105 | Val Loss: 0.0462
Train Acc: 99.68% | Val Acc: 98.31%
Epoch Time: 1.00 minutes

Seed 52 | Epoch 8/10
Train Loss: 0.0097 | Val Loss: 0.0460
Train Acc: 99.82% | Val Acc: 98.31%
Epoch Time: 1.01 minutes

Seed 52 | Epoch 9/10
Train Loss: 0.0090 | Val Loss: 0.0461

Train Acc: 99.78% | Val Acc: 98.31%
Epoch Time: 1.00 minutes

Seed 52 | Epoch 10/10
Train Loss: 0.0109 | Val Loss: 0.0464
Train Acc: 99.70% | Val Acc: 98.43%
Epoch Time: 1.00 minutes

=====

SEED 52 TRAINING COMPLETE
Time taken: 10.01 minutes

=====

Calculating final metrics for Seed 52...

=====

SEED 52 - FINAL PERFORMANCE

=====

Metric	Training	Validation	Testing
Accuracy (%)	99.8918	98.3112	97.2332
Sensitivity (%)	99.9353	98.5772	97.3881
Specificity (%)	99.8480	97.9228	97.0588
Precision (%)	99.8493	98.5772	97.3881
F1-Score (%)	99.8923	98.5772	97.3881
Error Rate (%)	0.1082	1.6888	2.7668
ROC-AUC	1.0000	0.9989	0.9992

Seed 52 - Training Confusion Matrix:
[[4599 7]
[3 4637]]

Seed 52 - Validation Confusion Matrix:
[[330 7]
[7 485]]

Seed 52 - Testing Confusion Matrix:
[[231 7]
[7 261]]

Completed 2/5 seeds.

=====

STARTING SEED 62

=====

Seed 62 | Epoch 1/10
Train Loss: 0.2861 | Val Loss: 0.1197
Train Acc: 89.22% | Val Acc: 95.54%
Epoch Time: 0.99 minutes

Seed 62 | Epoch 2/10
Train Loss: 0.0431 | Val Loss: 0.0688
Train Acc: 98.72% | Val Acc: 96.74%
Epoch Time: 1.00 minutes

Seed 62 | Epoch 3/10
Train Loss: 0.0202 | Val Loss: 0.0496
Train Acc: 99.43% | Val Acc: 97.71%
Epoch Time: 0.99 minutes

Seed 62 | Epoch 4/10
Train Loss: 0.0156 | Val Loss: 0.0443
Train Acc: 99.57% | Val Acc: 98.31%
Epoch Time: 0.98 minutes

Seed 62 | Epoch 5/10
Train Loss: 0.0138 | Val Loss: 0.0412
Train Acc: 99.60% | Val Acc: 98.91%
Epoch Time: 0.97 minutes

Seed 62 | Epoch 6/10
Train Loss: 0.0104 | Val Loss: 0.0379
Train Acc: 99.78% | Val Acc: 98.67%
Epoch Time: 0.98 minutes

Seed 62 | Epoch 7/10
Train Loss: 0.0111 | Val Loss: 0.0387
Train Acc: 99.73% | Val Acc: 98.43%
Epoch Time: 0.99 minutes

Seed 62 | Epoch 8/10
Train Loss: 0.0101 | Val Loss: 0.0383
Train Acc: 99.72% | Val Acc: 98.91%
Epoch Time: 1.00 minutes

Seed 62 | Epoch 9/10
Train Loss: 0.0112 | Val Loss: 0.0389
Train Acc: 99.71% | Val Acc: 98.43%
Epoch Time: 0.99 minutes

Seed 62 | Epoch 10/10
Train Loss: 0.0125 | Val Loss: 0.0386
Train Acc: 99.64% | Val Acc: 98.67%
Epoch Time: 1.00 minutes

=====

SEED 62 TRAINING COMPLETE

Time taken: 9.89 minutes

=====

Calculating final metrics for Seed 62...

=====

SEED 62 - FINAL PERFORMANCE

=====

Metric	Training	Validation	Testing
Accuracy (%)	99.8810	98.6731	98.2213
Sensitivity (%)	99.8491	99.3902	98.8806
Specificity (%)	99.9132	97.6261	97.4790
Precision (%)	99.9137	98.3903	97.7860
F1-Score (%)	99.8814	98.8878	98.3302
Error Rate (%)	0.1190	1.3269	1.7787
ROC-AUC	1.0000	0.9994	0.9983

Seed 62 - Training Confusion Matrix:

```
[[4602  4]
 [  7 4633]]
```

Seed 62 - Validation Confusion Matrix:

```
[[329  8]
 [  3 489]]
```

Seed 62 - Testing Confusion Matrix:

```
[[232  6]
 [  3 265]]
```

Completed 3/5 seeds.

=====

STARTING SEED 72

=====

Seed 72 | Epoch 1/10
Train Loss: 0.2909 | Val Loss: 0.1359
Train Acc: 88.18% | Val Acc: 94.93%
Epoch Time: 1.00 minutes

Seed 72 | Epoch 2/10
Train Loss: 0.0448 | Val Loss: 0.0761
Train Acc: 98.69% | Val Acc: 97.47%
Epoch Time: 0.99 minutes

Seed 72 | Epoch 3/10
Train Loss: 0.0257 | Val Loss: 0.0526
Train Acc: 99.24% | Val Acc: 98.67%
Epoch Time: 0.98 minutes

Seed 72 | Epoch 4/10

Train Loss: 0.0174 | Val Loss: 0.0501
Train Acc: 99.51% | Val Acc: 98.55%
Epoch Time: 0.97 minutes

Seed 72 | Epoch 5/10
Train Loss: 0.0155 | Val Loss: 0.0465
Train Acc: 99.49% | Val Acc: 98.55%
Epoch Time: 0.97 minutes

Seed 72 | Epoch 6/10
Train Loss: 0.0134 | Val Loss: 0.0446
Train Acc: 99.60% | Val Acc: 98.91%
Epoch Time: 0.97 minutes

Seed 72 | Epoch 7/10
Train Loss: 0.0114 | Val Loss: 0.0451
Train Acc: 99.74% | Val Acc: 98.79%
Epoch Time: 1.00 minutes

Seed 72 | Epoch 8/10
Train Loss: 0.0111 | Val Loss: 0.0464
Train Acc: 99.71% | Val Acc: 98.55%
Epoch Time: 1.00 minutes

Seed 72 | Epoch 9/10
Train Loss: 0.0121 | Val Loss: 0.0417
Train Acc: 99.69% | Val Acc: 98.91%
Epoch Time: 1.01 minutes

Seed 72 | Epoch 10/10
Train Loss: 0.0129 | Val Loss: 0.0442
Train Acc: 99.68% | Val Acc: 98.91%
Epoch Time: 1.00 minutes

=====

SEED 72 TRAINING COMPLETE

Time taken: 9.90 minutes

=====

Calculating final metrics for Seed 72...

=====

SEED 72 - FINAL PERFORMANCE			
Metric	Training	Validation	Testing
Accuracy (%)	99.8810	98.9144	98.0237
Sensitivity (%)	99.8491	99.3902	98.8806
Specificity (%)	99.9132	98.2196	97.0588
Precision (%)	99.9137	98.7879	97.4265
F1-Score (%)	99.8814	99.0881	98.1481
Error Rate (%)	0.1190	1.0856	1.9763
ROC-AUC	1.0000	0.9994	0.9983

=====

Seed 72 - Training Confusion Matrix:

```
[[4602  4]
 [  7 4633]]
```

Seed 72 - Validation Confusion Matrix:

```
[[331  6]
 [  3 489]]
```

Seed 72 - Testing Confusion Matrix:

```
[[231  7]
 [  3 265]]
```

Completed 4/5 seeds.

=====

STARTING SEED 82

=====

Seed 82 | Epoch 1/10
Train Loss: 0.3011 | Val Loss: 0.1217
Train Acc: 88.42% | Val Acc: 95.17%

Epoch Time: 0.98 minutes

Seed 82 | Epoch 2/10
Train Loss: 0.0417 | Val Loss: 0.0643
Train Acc: 98.92% | Val Acc: 97.83%
Epoch Time: 0.99 minutes

Seed 82 | Epoch 3/10
Train Loss: 0.0209 | Val Loss: 0.0407
Train Acc: 99.49% | Val Acc: 98.55%
Epoch Time: 0.97 minutes

Seed 82 | Epoch 4/10
Train Loss: 0.0148 | Val Loss: 0.0432
Train Acc: 99.56% | Val Acc: 98.43%
Epoch Time: 0.96 minutes

Seed 82 | Epoch 5/10
Train Loss: 0.0106 | Val Loss: 0.0412
Train Acc: 99.71% | Val Acc: 98.43%
Epoch Time: 0.98 minutes

Seed 82 | Epoch 6/10
Train Loss: 0.0107 | Val Loss: 0.0377
Train Acc: 99.71% | Val Acc: 98.55%
Epoch Time: 0.98 minutes

Seed 82 | Epoch 7/10
Train Loss: 0.0111 | Val Loss: 0.0373
Train Acc: 99.73% | Val Acc: 98.43%
Epoch Time: 1.00 minutes

Seed 82 | Epoch 8/10
Train Loss: 0.0119 | Val Loss: 0.0411
Train Acc: 99.63% | Val Acc: 98.19%
Epoch Time: 0.99 minutes

Seed 82 | Epoch 9/10
Train Loss: 0.0111 | Val Loss: 0.0363
Train Acc: 99.71% | Val Acc: 98.67%
Epoch Time: 0.98 minutes

Seed 82 | Epoch 10/10
Train Loss: 0.0117 | Val Loss: 0.0369
Train Acc: 99.68% | Val Acc: 98.67%
Epoch Time: 0.97 minutes

=====

SEED 82 TRAINING COMPLETE

Time taken: 9.81 minutes

=====

Calculating final metrics for Seed 82...

=====

SEED 82 - FINAL PERFORMANCE			
Metric	Training	Validation	Testing
Accuracy (%)	99.8810	98.6731	98.0237
Sensitivity (%)	99.8922	99.3902	98.8806
Specificity (%)	99.8697	97.6261	97.0588
Precision (%)	99.8707	98.3903	97.4265
F1-Score (%)	99.8815	98.8878	98.1481
Error Rate (%)	0.1190	1.3269	1.9763
ROC-AUC	1.0000	0.9994	0.9971

=====

Seed 82 - Training Confusion Matrix:

```
[[4600  6]
 [  5 4635]]
```

Seed 82 - Validation Confusion Matrix:

```
[[329  8]
 [  3 489]]
```

Seed 82 - Testing Confusion Matrix:

```
[[231  7]
 [  3 265]]
```

Completed 5/5 seeds.

PER-SEED RESULTS

Seed	Dataset	Metric	Value
42	Training	Accuracy (%)	99.9135
42	Validation	Accuracy (%)	98.3112
42	Testing	Accuracy (%)	97.6285
42	Training	Sensitivity (%)	99.9353
42	Validation	Sensitivity (%)	98.7805
42	Testing	Sensitivity (%)	97.7612
42	Training	Specificity (%)	99.8914
42	Validation	Specificity (%)	97.6261
42	Testing	Specificity (%)	97.4790
42	Training	Precision (%)	99.8923
42	Validation	Precision (%)	98.3806
42	Testing	Precision (%)	97.7612
42	Training	F1-Score (%)	99.9138
42	Validation	F1-Score (%)	98.5801
42	Testing	F1-Score (%)	97.7612
42	Training	Error Rate (%)	0.0865
42	Validation	Error Rate (%)	1.6888
42	Testing	Error Rate (%)	2.3715
42	Training	ROC-AUC	1.0000
42	Validation	ROC-AUC	0.9991
42	Testing	ROC-AUC	0.9987
52	Training	Accuracy (%)	99.8918
52	Validation	Accuracy (%)	98.3112
52	Testing	Accuracy (%)	97.2332
52	Training	Sensitivity (%)	99.9353
52	Validation	Sensitivity (%)	98.5772
52	Testing	Sensitivity (%)	97.3881
52	Training	Specificity (%)	99.8480
52	Validation	Specificity (%)	97.9228
52	Testing	Specificity (%)	97.0588
52	Training	Precision (%)	99.8493
52	Validation	Precision (%)	98.5772
52	Testing	Precision (%)	97.3881
52	Training	F1-Score (%)	99.8923
52	Validation	F1-Score (%)	98.5772
52	Testing	F1-Score (%)	97.3881
52	Training	Error Rate (%)	0.1082
52	Validation	Error Rate (%)	1.6888
52	Testing	Error Rate (%)	2.7668
52	Training	ROC-AUC	1.0000
52	Validation	ROC-AUC	0.9989
52	Testing	ROC-AUC	0.9992
62	Training	Accuracy (%)	99.8810
62	Validation	Accuracy (%)	98.6731
62	Testing	Accuracy (%)	98.2213
62	Training	Sensitivity (%)	99.8491
62	Validation	Sensitivity (%)	99.3902
62	Testing	Sensitivity (%)	98.8806
62	Training	Specificity (%)	99.9132
62	Validation	Specificity (%)	97.6261
62	Testing	Specificity (%)	97.4790
62	Training	Precision (%)	99.9137
62	Validation	Precision (%)	98.3903
62	Testing	Precision (%)	97.7860
62	Training	F1-Score (%)	99.8814
62	Validation	F1-Score (%)	98.8878
62	Testing	F1-Score (%)	98.3302
62	Training	Error Rate (%)	0.1190
62	Validation	Error Rate (%)	1.3269
62	Testing	Error Rate (%)	1.7787
62	Training	ROC-AUC	1.0000
62	Validation	ROC-AUC	0.9994
62	Testing	ROC-AUC	0.9983
72	Training	Accuracy (%)	99.8810
72	Validation	Accuracy (%)	98.9144
72	Testing	Accuracy (%)	98.0237
72	Training	Sensitivity (%)	99.8491

```

72 Validation Sensitivity (%) 99.3902
72 Testing Sensitivity (%) 98.8806
72 Training Specificity (%) 99.9132
72 Validation Specificity (%) 98.2196
72 Testing Specificity (%) 97.0588
72 Training Precision (%) 99.9137
72 Validation Precision (%) 98.7879
72 Testing Precision (%) 97.4265
72 Training F1-Score (%) 99.8814
72 Validation F1-Score (%) 99.0881
72 Testing F1-Score (%) 98.1481
72 Training Error Rate (%) 0.1190
72 Validation Error Rate (%) 1.0856
72 Testing Error Rate (%) 1.9763
72 Training ROC-AUC 1.0000
72 Validation ROC-AUC 0.9994
72 Testing ROC-AUC 0.9983
82 Training Accuracy (%) 99.8810
82 Validation Accuracy (%) 98.6731
82 Testing Accuracy (%) 98.0237
82 Training Sensitivity (%) 99.8922
82 Validation Sensitivity (%) 99.3902
82 Testing Sensitivity (%) 98.8806
82 Training Specificity (%) 99.8697
82 Validation Specificity (%) 97.6261
82 Testing Specificity (%) 97.0588
82 Training Precision (%) 99.8707
82 Validation Precision (%) 98.3903
82 Testing Precision (%) 97.4265
82 Training F1-Score (%) 99.8815
82 Validation F1-Score (%) 98.8878
82 Testing F1-Score (%) 98.1481
82 Training Error Rate (%) 0.1190
82 Validation Error Rate (%) 1.3269
82 Testing Error Rate (%) 1.9763
82 Training ROC-AUC 1.0000
82 Validation ROC-AUC 0.9994
82 Testing ROC-AUC 0.9971

```

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MEAN ± SD PERFORMANCE

=====

Dataset	Metric	Mean	SD	Mean ± SD
Training	Accuracy (%)	99.889682	0.014102	99.89 ± 0.01
Training	Sensitivity (%)	99.892241	0.043103	99.89 ± 0.04
Training	Specificity (%)	99.887104	0.028307	99.89 ± 0.03
Training	Precision (%)	99.887950	0.028027	99.89 ± 0.03
Training	F1-Score (%)	99.890086	0.014069	99.89 ± 0.01
Training	Error Rate (%)	0.110318	0.014102	0.11 ± 0.01
Training	ROC-AUC	0.999995	0.000002	1.00 ± 0.00
Validation	Accuracy (%)	98.576598	0.261514	98.58 ± 0.26
Validation	Sensitivity (%)	99.105691	0.396211	99.11 ± 0.40
Validation	Specificity (%)	97.804154	0.265409	97.80 ± 0.27
Validation	Precision (%)	98.505273	0.178196	98.51 ± 0.18
Validation	F1-Score (%)	98.804207	0.221538	98.80 ± 0.22
Validation	Error Rate (%)	1.423402	0.261514	1.42 ± 0.26
Validation	ROC-AUC	0.999262	0.000229	1.00 ± 0.00
Testing	Accuracy (%)	97.826087	0.395257	97.83 ± 0.40
Testing	Sensitivity (%)	98.358209	0.727373	98.36 ± 0.73
Testing	Specificity (%)	97.226891	0.230136	97.23 ± 0.23
Testing	Precision (%)	97.557635	0.197952	97.56 ± 0.20
Testing	F1-Score (%)	97.955158	0.378946	97.96 ± 0.38
Testing	Error Rate (%)	2.173913	0.395257	2.17 ± 0.40
Testing	ROC-AUC	0.998332	0.000784	1.00 ± 0.00

=====

TRAINING / VALIDATION / TESTING

MEAN ± SD PERFORMANCE COMPARISON TABLE

=====

Metric	Training Mean ± SD	Validation Mean ± SD	Testing Mean ± SD
Accuracy (%)	99.89 ± 0.01	98.58 ± 0.26	97.83 ± 0.40
Sensitivity (%)	99.89 ± 0.04	99.11 ± 0.40	98.36 ± 0.73
Specificity (%)	99.89 ± 0.03	97.80 ± 0.27	97.23 ± 0.23
Precision (%)	99.89 ± 0.03	98.51 ± 0.18	97.56 ± 0.20
F1-Score (%)	99.89 ± 0.01	98.80 ± 0.22	97.96 ± 0.38

Error Rate (%)	0.11 ± 0.01	1.42 ± 0.26	2.17 ± 0.40
ROC-AUC	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00

=====

PER-SEED TRAINING TIME

=====

Seed	Time (minutes)	Time (hours)
42	12.50	0.21
52	10.01	0.17
62	9.89	0.16
72	9.90	0.17
82	9.81	0.16

Mean Time: 10.42 minutes
SD Time: 1.16 minutes
Mean ± SD Time: 10.42 ± 1.16 minutes

Total Experiment Time: 0.98 hours

=====

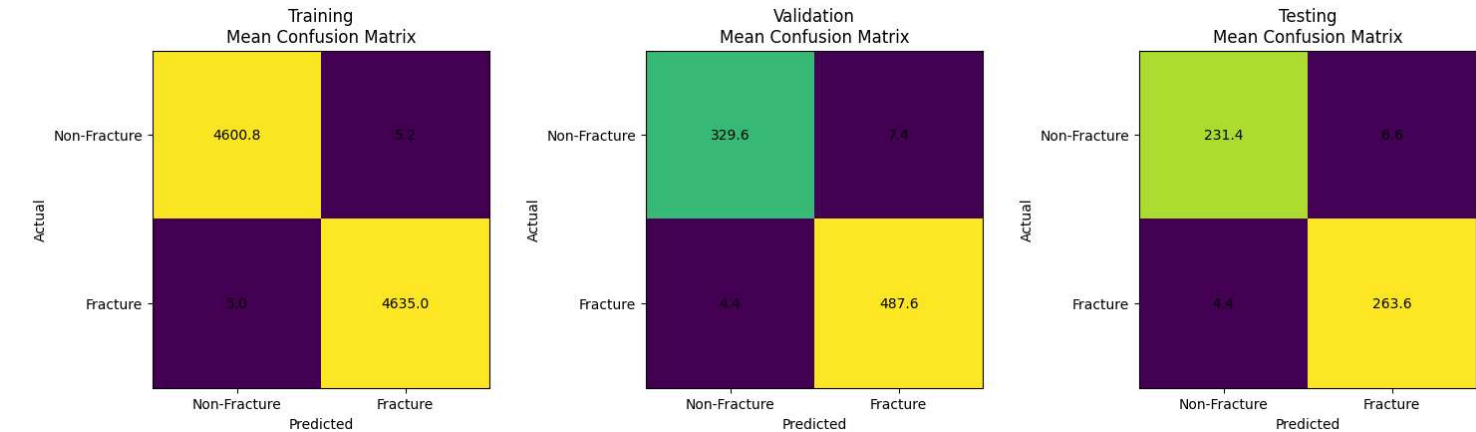
MEAN CONFUSION MATRICES

=====

Training Mean CM:
[[4600.8 5.2]
[5. 4635.]]

Validation Mean CM:
[[329.6 7.4]
[4.4 487.6]]

Testing Mean CM:
[[231.4 6.6]
[4.4 263.6]]

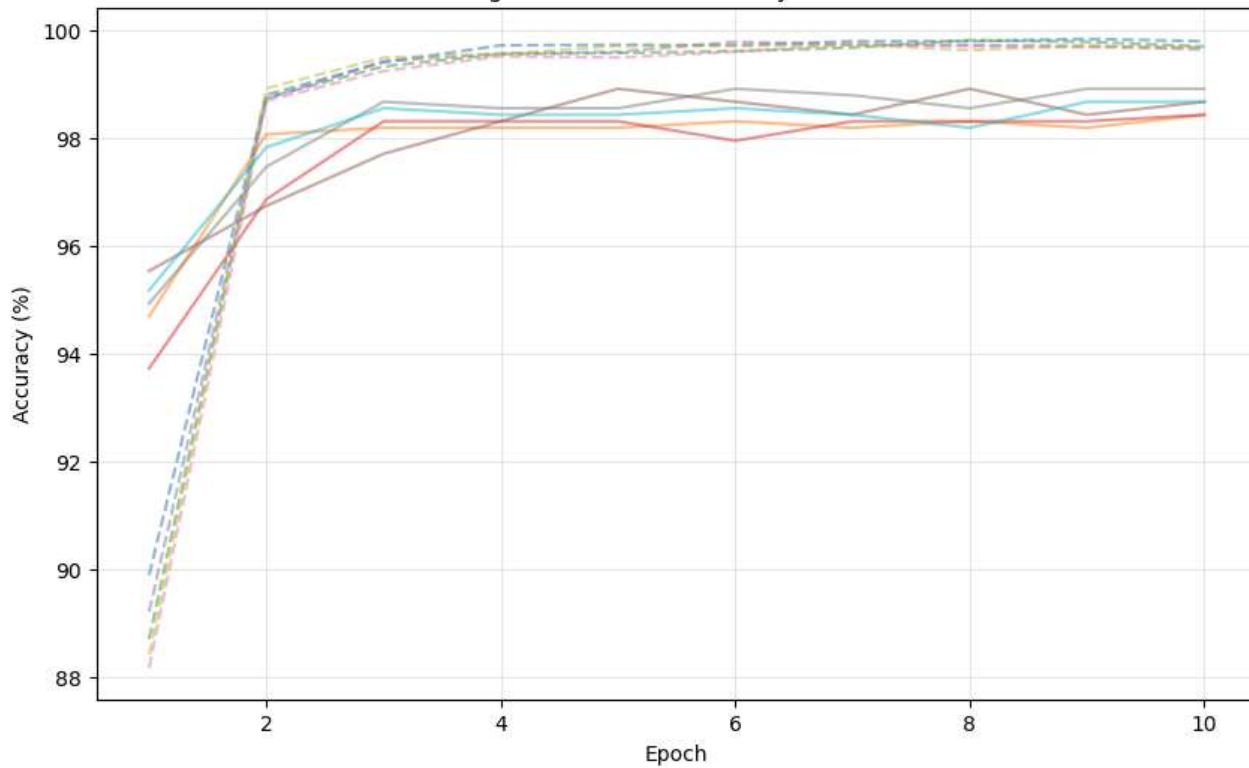
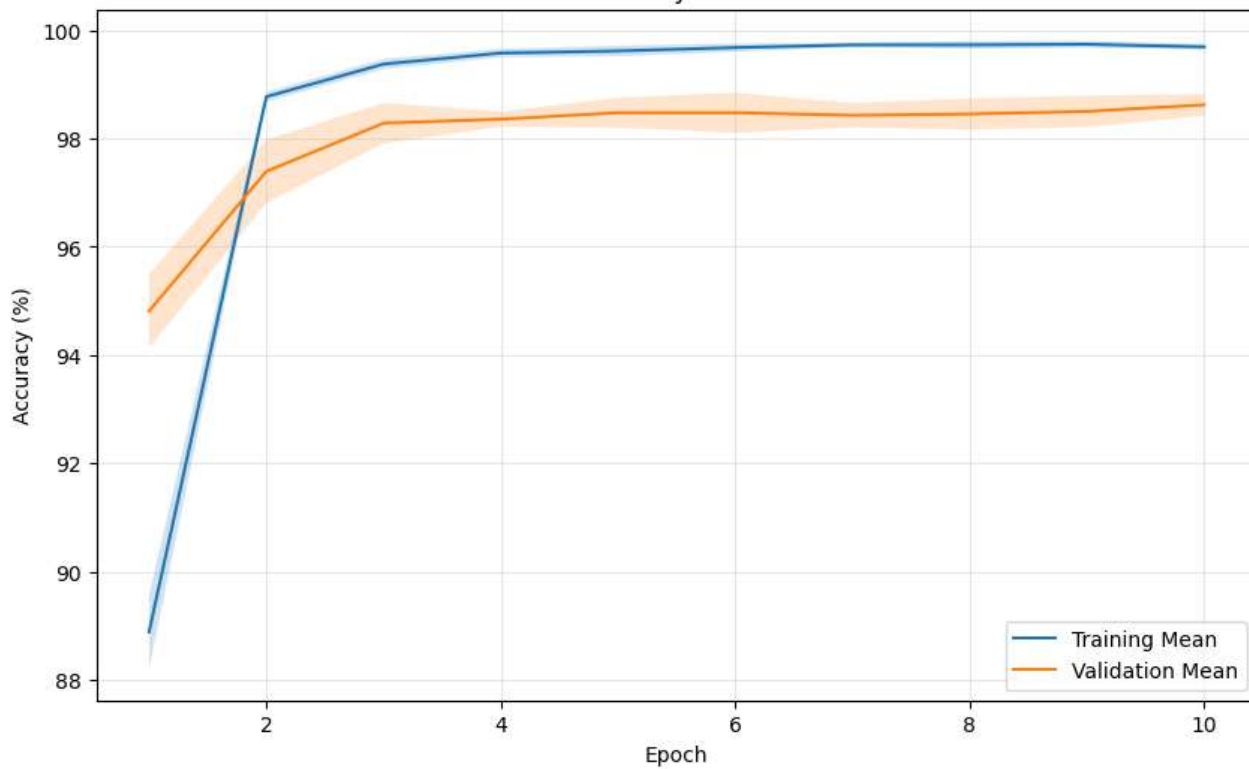


Training Confusion Matrix Mean ± SD:
['4600.80 ± 1.30', '5.20 ± 1.30']
['5.00 ± 2.00', '4635.00 ± 2.00']

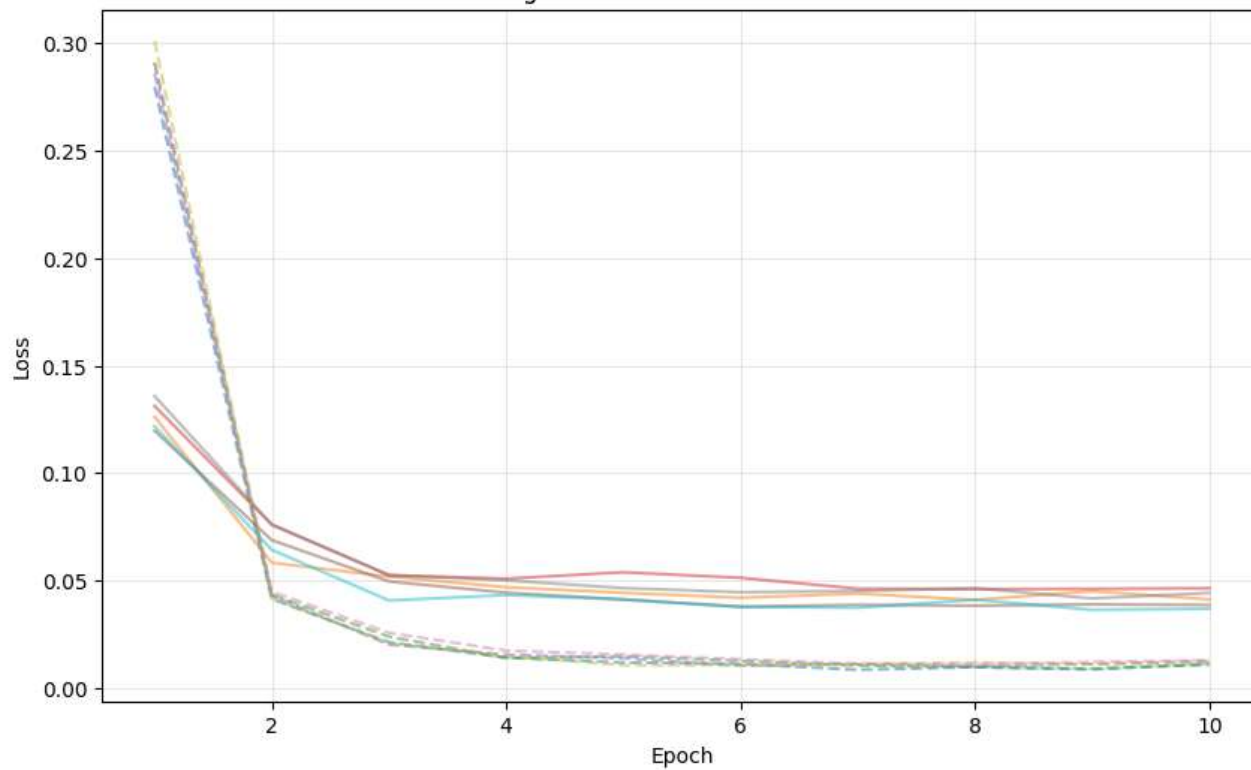
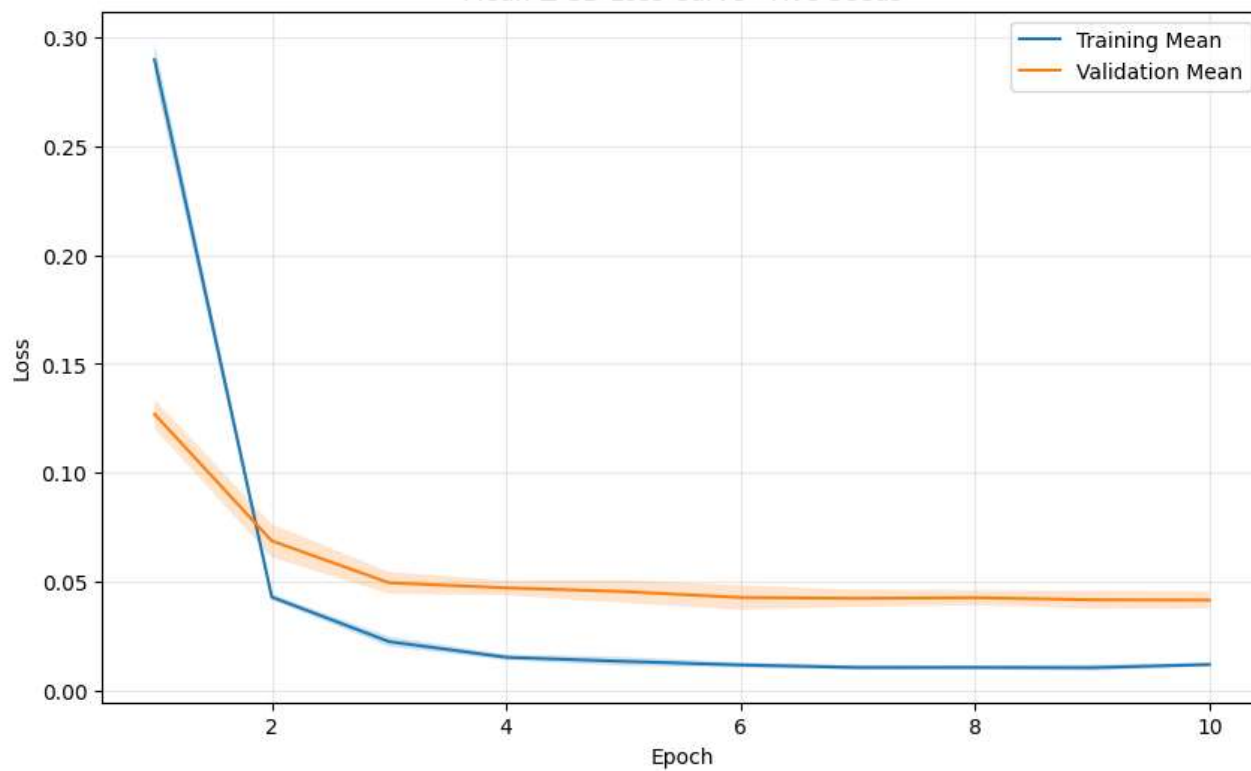
Validation Confusion Matrix Mean ± SD:
['329.60 ± 0.89', '7.40 ± 0.89']
['4.40 ± 1.95', '487.60 ± 1.95']

Testing Confusion Matrix Mean ± SD:
['231.40 ± 0.55', '6.60 ± 0.55']
['4.40 ± 1.95', '263.60 ± 1.95']

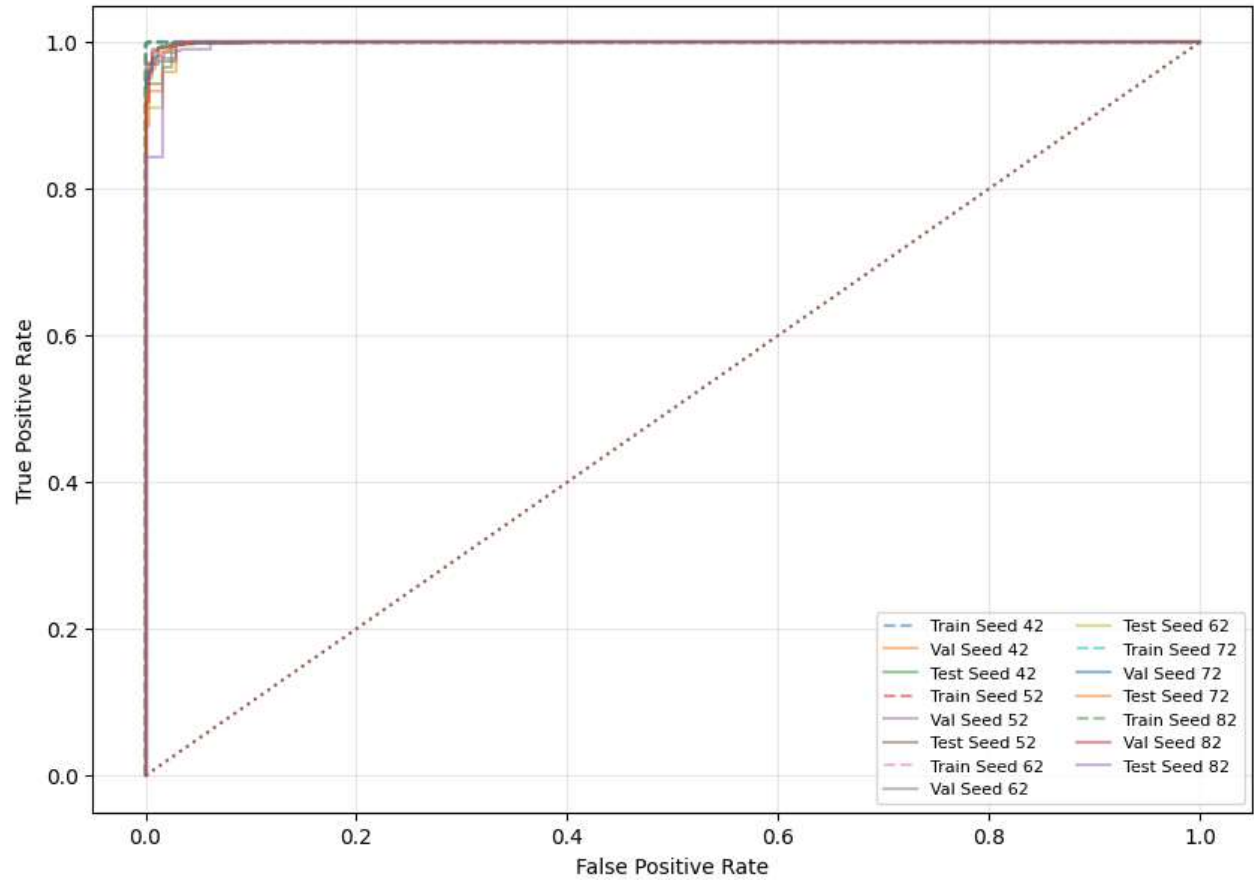
Training vs Validation Accuracy - Five Seeds

Mean $\pm$ SD Accuracy Curve - Five Seeds

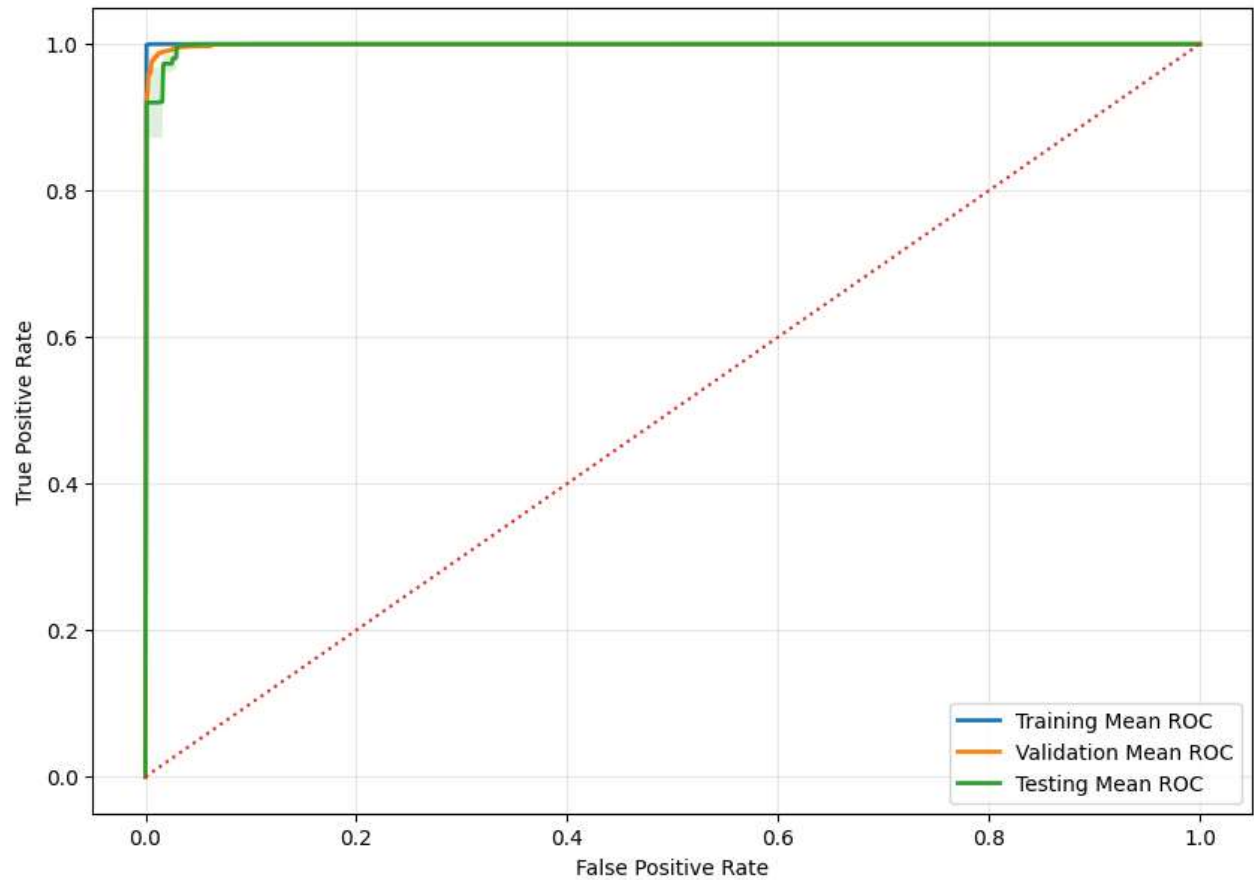
Training vs Validation Loss - Five Seeds

Mean $\pm$ SD Loss Curve - Five Seeds

ROC Curves - Five Seeds



Mean $\pm$ SD ROC Curves - Five Seeds



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FINAL EXPERIMENT SUMMARY

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Architecture:
EfficientNet-B0 → 1280 features → Dropout(0.2) → Linear(1280→1) → Sigmoid (evaluation)

Seeds: [42, 52, 62, 72, 82]
Epochs per seed: 10
Batch size: 64
Learning rate: 0.0001

Total experiment time: 0.98 hours

Mean ± SD results are shown above.

Experiment completed successfully.
```